Mateo Díaz

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Research Interests

I am interested in the interplay between optimization, geometry, and statistics, and its applications to data science, learning, and signal processing.

Education

Cornell University

Ph.D. Applied Mathematics (2016−2021)
Committee: Damek Davis (Advisor), Robert Kleinberg, Adrian Lewis, and James Renegar
Center for Applied Mathematics;

M.S. Computer Science (2016−2018)
Computer Science Department.

Universidad de los Andes

M.S. Mathematics (2014−2016)
B.S. Mathematics (2010−2013)
Department of Mathematics;

B.S. Systems and Computing Engineering (2010−2015)
Systems and Computing Engineering Department.

Employment history

University of Chicago
Department of Statistics
Faraco Family Assistant Professor of Data Science (October 2026 −)
Assistant Professor (July 2026 − August 2026)

Johns Hopkins University
Department of Applied Mathematics and Statistics
Assistant Professor (July 2023−June 2026)

California Institute of Technology
Computing + Mathematical Sciences
Postdoctoral Scholar (September 2021−June 2023). Hosts: Venkat Chandrasekaran and Joel Tropp.

Google Research
Research Intern (Fall 2020). Hosts: Miles Lubin and David Applegate.

Rappi (Colombia)
Lead Research Scientist (Summer 2020).

Google Maps
Software Intern (Summer 2019).

wOzy
Co-founder and developer (2010-2015).
I co-founded a video game startup during my undergraduate years.

Publications

Preprints

Benign population landscapes do not transfer to empirical risk minimization in nonsmooth optimization
(with P. Wang and L. Jiang)
Submitted, 2026.

Wasserstein approximation of measures via orthogonal frames: statistical and computational complexities
(with N. García Trillos, D. López-Castaño, and C. Yang)
Submitted, 2026.

Nonsmooth Riemannian optimization with inexact manifold primitives via bundle methods
(with B. Grimmer and I. McPherson)
Submitted, 2026.

Active set identification and rapid convergence for degenerate primal-dual problems
(with P. Izquierdo Lehmann, H. Lu, J. Yang)
Submitted, 2026.

Preconditioned subgradient method for composite optimization: overparameterization and fast convergence
(with L. Jiang, A. G. Labassi)
Submitted, 2025.

Controlling the False Discovery Rate in Subspace Selection
(with V. Chandrasekaran)
Submitted, 2024.

Published

Any-Dimensional Invariant Universality
(with S. Yao and E. Levin)
ICML, 2026.

μpscaling Small Models: Principled Warm Starts and Hyperparameter Transfer
(with Y. Ma, N. Chen, S. Hayou, D. Kunisky, S. Villar)
ICML, 2026.

Invariant Kernels: Rank Stabilization and Generalization Across Dimensions
(with D. Drusvyatskiy, J. Kendrick, R. R. Thomas)
Information and Inference (To appear), 2026.

Robust, randomized preconditioning for kernel ridge regression
(with E. Epperly, Z. Frangella, J. Tropp, and R. Webber)
Advances in Computational Mathematics (To appear), 2026.

PDLP: A Practical First-Order Method for Large-Scale Linear Programming
(with D. Applegate, O. Hinder, H. Lu, M. Lubin, B. O'Donoghue, and W. Schudy)
Mathematical Programming Computation, 2026.

The radius of statistical efficiency
(with J. Cutler and D. Drusvyatskiy)
Foundations of Computational Mathematics, 2026.

Beyond Scores: Proximal Diffusion Models
(with Z. Fang, S. Buchanan, J. Sulam)
NeurIPS, 2025.
Research Spotlight, *CPAL*, 2026.

On Transferring Transferability: Towards a Theory for Size Generalization
(with E. Levin, Y. Ma, S. Villar)
NeurIPS (**Spotlight**), 2025.

Clustering a Mixture of Gaussians with Unknown Covariance
(with D. Davis and K. Wang)
Bernoulli, 2025.

Stochastic approximation with decision-dependent distributions: asymptotic normality and optimality
(with J. Cutler and D. Drusvyatskiy)
Journal of Machine Learning Research, 2024.

Any-dimensional equivariant neural networks
(with E. Levin)
AISTATS, 2024.

Infeasibility detection with primal-dual hybrid gradient for large-scale linear programs
(with D. Applegate, H. Lu, and M. Lubin)
SIAM Journal on Optimization, 2024.

Optimal Convergence Rates for the Proximal Bundle Method
(with B. Grimmer)
SIAM Journal on Optimization, 2023.

Escaping strict saddle points of the Moreau envelope in nonsmooth optimization
(with D. Davis and D. Drusvyatskiy)
SIAM Journal on Optimization, 2022.

Optimization of vaccination for COVID-19 in the midst of a pandemic
(with Q. Luo, R. Weightman, S. T. McQuade, E. Trélat, W. Barbour, D. Work, S. Samanaranayake, B. Piccoli)
Networks and Heterogeneous Media, 2022.

Low-rank matrix recovery with composite optimization: good conditioning and rapid convergence
(with V. Charisopoulos, Y. Chen, D. Davis, L. Ding, D. Drusvyatskiy)
Foundations of Computational Mathematics, 2021.

Practical Large-Scale Linear Programming using Primal-Dual Hybrid Gradient
(with D. Applegate, O. Hinder, H. Lu, M. Lubin, B. O'Donoghue, and W. Schudy),
NeurIPS, 2021. **Beale — Orchard-Hays Prize**.

Composite optimization for robust blind deconvolution
(with V. Charisopoulos, D. Davis, D. Drusvyatskiy)
Information and Inference, 2020.

Efficient Clustering for Stretched Mixtures: Landscape and Optimality
(with K. Wang and Y. Yan),
NeurIPS, 2020.

Local angles and dimension estimation from data on manifold
(with A. Quiroz, M. Velasco)
Journal of Multivariate Analysis, 2019.

The nonsmooth landscape of blind deconvolution
NeurIPS Workshop on Optimization for Machine Learning, 2019.

Compressed sensing of data with known distribution
(with M. Junca, F. Rincón, M. Velasco)
Applied and Computational Harmonic Analysis, 2018.

In search of balance: The challenge of generating balanced Latin rectangles
(with C. Gomes, R. Le Bras)
CPAIOR, 2017.

Grants and Awards

Sloan Research Fellowship in Mathematics, 2026.

NSF MFAI Award (PI): "Any-Dimensional Equivariant Learning," 2025.

NSF CAREER Award: "Scalable Optimization for Data Science: Complexity and Structure," 2025.

Mathematical Optimization Society: Beale — Orchard-Hays Prize, 2024.

Universidad de los Andes: Fellowship for Master's Degree, 2014-2016.

Ecopetrol: "Bachilleres por Colombia" Scholarship, 2010-2014.
Full scholarship for undergraduate studies given to the best ICFES score (SAT equivalent) in each State.

Students

Pedro Izquierdo Lehmann, Statistics PhD, University of Chicago (2026 -).

Abdel Ghani Labassi, AMS PhD, Johns Hopkins University (2022 -).

Daniel Lopez-Castaño, AMS PhD, Johns Hopkins University (2021 -), co-advised with Soledad Villar.

Ian McPherson, AMS PhD, Johns Hopkins University (2025 -), co-advised with Mauro Maggioni.

Jonas Elmerraji, AMS DEng, Johns Hopkins University (2024 -), co-advised with James C. Spall.

Shengtai Yao, AMS Masters (2025). *Next position:* PhD student at Stanford University.

Haoruo Zhang, AMS Masters (2025). *Next position:* PhD student at University of Illinois Urbana-Champaign.

Fengsheng Lin, Data Science Masters (2025). *Next position:* PhD student at University of Washington.

Yun You, Data Science Masters (2024).

Talks

FOLDS Seminar (University of Pennsylvania), Philadelphia, PA, Sep 2026.

Modeling and Optimization: Theory and Applications (MOPTA), Bethlehem, PA, Aug 2026.

International Conference on Optimization and Machine Learning, Theory and Applications (ICOMALTA), Arica, Chile, May 2026.

DSI/Statistics Colloquium (University of Chicago), Chicago, IL, Feb 2026.

Joint Mathematics Meetings (JMM), Washington, DC, Jan 2026. **Two talks**.

Optimization and ML Seminar (Universidad de la República), Montevideo, Uruguay, Dec 2025.

Workshop on Mathematical Analysis and Machine Learning (University of Pittsburgh), Pittsburgh, PA, Nov 2025.

Statistics Seminar (University of Maryland), College Park, MD, Nov 2025.

Nonlinear Analysis Seminar (North Carolina State University), Remote, Nov 2025.

INFORMS Annual Meeting, Atlanta, GA, Oct 2025.

IDeAS Seminar (Princeton University), Princeton, NJ, Sep 2025.

International Conference on Continuous Optimization (ICCOPT), Los Angeles, CA, Jul 2025.

INFORMS Applied Probability Society, Atlanta, GA, Jul 2025.

Modeling and Optimization: Theory and Applications (MOPTA), Ponta Delgada, Portugal, Jun 2025.

Portuguese American Workshop on Optimization (PAOW), Horta, Portugal, Jun 2025.

Workshop: Optimization and learning: theory and applications, Montreal, Canada, May 2025.

Optimization and Statistical Learning Workshop (Columbia University), New York, NY, April 2025.

Optimization Workshop: Theory, Algorithms, and Applications, Bogotá, Colombia, December 2024.

SIAM Mathematics of Data Science Conference, Atlanta, GA, October 2024.

INFORMS Annual Meeting, Seattle, WA, October 2024.

Applied Mathematics Seminar (University of Washington), Seattle, WA, October 2024.

26th International Symposium on Mathematical Theory of Networks and Systems, Cambridge, UK, August 2024.

11th World Congress in Probability and Statistics, Bochum, Germany, August 2024.

International Symposium on Mathematical Programming, Montreal, Canada, July 2024.

INFORMS International, Medellín, Colombia, June 2024.

3rd Colombian Conference on Applied and Industrial Mathematics, Bucaramanga, Colombia, June 2024.

Artificial Intelligence and Statistics, Valencia, Spain, May 2024. **Poster**.

Industrial and Systems Engineering Seminar (Lehigh University), Bethlehem, PA, April 2024.

Statistics Institute Seminar (Universidad de la República), Montevideo, Uruguay, March 2024.

SIAM Parallel Processing for Scientific Computing, Baltimore, MD, March 2024.

Algebra and Cryptography Seminar (Stevens Institute), Hoboken, NJ, February 2024.

Operations Research Seminar at the Tepper School of Business (Carnegie Mellon University), Pittsburgh, PA, December 2023.

Data Science Week, Purdue University, Fort Wayne, IN, November 2023. **Keynote Speaker**.

Jornadas de Probabilidad (Universidad Nacional de Colombia), Bogotá, Colombia, November 2023.

INFORMS Annual Meeting, Phoenix, AZ, October 2023.

International Congress on Industrial and Applied Mathematics, Tokyo, Japan, August 2023.

Modeling and Optimization: Theory and Applications, Bethlehem, PA, August 2023

SIAM Conference on Optimization, Seattle, WA, June 2023.

Workshop at University of Washington, Seattle, WA, June 2023.

Statistics Seminar (University of Wisconsin-Madison), Madison, WI, April 2023.

Probability and Statistics Seminar (University of Southern California), Los Angeles, CA, February 2023.

US-Mexico Workshop on Optimization and Applications, Huatulco, Mexico January 2023.

Workshop on Scientific Machine Learning, Santiago, Chile November 2022.

Quantitative Methods seminar at Krannert School of Management (Purdue University), West Lafayette, IN October 2022.

INFORMS Annual Meeting, Indianapolis, IN October 2022.

SIAM Conference on Mathematics of Data Science, San Diego, CA September 2022.

International Conference on Continuous Optimization, Bethlehem, PA July 2022.

2nd Colombian Congress of Applied and Industrial Mathematics (MAPI 2), Medellin, Colombia June 2022.

Workshop on Robustness and Resilience in Stochastic Optimization and Statistical Learning, Erice, Italy May 2022.

INFORMS Optimization Society Conference, Greenville, SC March 2022.

Seminario de Estadística, Control y Optimización (Universidad de los Andes), Bogotá Colombia 2021.

Combinatorics and Probability at UC Irvine, Irvine, CA November 2021.

CMX Seminar at Caltech, Pasadena, CA October 2021.

SIAM Conference on Optimization, Virtual July 2021.

PhD Defense at Cornell University, Ithaca, NY July 2021.

2021 MINDS Symposium on the Foundations of Data Science, Baltimore, MD January 2021.

Google Research, New York, NY January 2021.

SIAM Conference on Optimization, Hong Kong, China May 2020. (**Cancelled**)

SIAM Conference on Mathematics of Data Science, Cincinnati, OH May 2020. (**Cancelled**)

INFORMS Optimization Society Conference, Greenville, SC. March 2020. (**Postponed**)

RPI Applied Math days, Troy NY, April 2019.

Young Researchers Workshop (ORIE Cornell), Ithaca NY October 2017. Poster presenter.

Foundations of Computational Mathematics conference, Barcelona Spain June 2017. Poster Presenter.

Octava Escuela de Física Matemática, Bogotá Colombia May 2016.

The 22nd International Symposium on Mathematical Programming, Pittsburgh PA June 2015. Contributed Sessions.

Foundations of Computational Mathematics conference, Montevideo Uruguay December 2014. Poster Presenter.

Service

Department

Thesis committee member: Wilson Gregory (2026), Pedro Radamacher (Masters Universidad Catolica de Chile 2025), Shihong Wei (PhD JHU-AMS 2024), Diego Fernando Fonseca (PhD Uniandes 2023), and Diego Arevalo Ovalle (Masters Uniandes 2022).

DSI Hiring Committee (2026-2027).

AMS Hiring Committee (2025-2026).

AMS Data Science Masters Admission Committee (2024).

AMS Postdoc Hiring Committee (2024).

Event organization

Conference organizing committee: DeepMath 2024, Optimization Workshop: Theory, Computation, and Applications 2024 (Colombia); and Statistics and Data Science Workshop 2025 (Colombia).

Cluster chair: INFORMS International 2024; and Third Colombian Conference on Applied and Industrial Mathematics 2024.

Minisymposium organizer: INFORMS Optimization Society 2026; ICCOPT 2025; World Congress in Probability and Statistics 2024; ISMP 2024; INFORMS 2023; MOPTA 2023; SIAM Conference on Optimization 2023; SIAM Mathematics of Data Science 2022; INFORMS 2022; ICCOPT 2022; and INFORMS Optimization Society 2020 (Cancelled).

Seminar organizer: AMS Seminar at Johns Hopkins 2024-2025; CMI Student/Postdoc seminar at Caltech 2021-2023; and Optimization seminar at Cornell 2018-2020.

Reviewer

Grants: NSF (panelist) 2026, 2025, 2023; Department of Energy (reviewer) 2024.

Journals and Conferences: Foundations of Computational Mathematics, Mathematical Programming, SIAM Journal on Optimization, Mathematics of Operations Research, INFORMS Journal on Optimization, INFORMS Journal on Computing, Journal of Machine Learning Research, NeurIPS, Symposium on Theory of Computing (STOC), IEEE International Symposium on Information Theory, IEEE Transactions on Signal Processing, IEEE Transactions on Information Theory, and Optimization and Engineering.

Teaching

At **University of Chicago**:

CAAM/DATA/STAT 31019 Optimization for Data Science, Autumn 2026.

At **Johns Hopkins University**:

AMS 747 Mathematics of Data Science, Fall 2025,

AMS 762 Nonlinear Optimization II, Spring 2025,

AMS 761 Nonlinear Optimization I, Fall 2023, Fall 2024,

AMS 721 Probability Theory II, Spring 2024.

At **Cornell**:

ORIE 6340 Mathematics of Data Science, Spring 2021 (TA),

ORIE 5270 Big Data Technologies, Spring 2020,

ORIE 6125 Computational Methods in Operations Research, Spring 2020,

ORIE 3300 Optimization I, Summer 2017 (TA).

At **Universidad de los Andes**:

MATE-1201 Precalculus, Spring 2015 (TA),

MATE-2604 Numerical Analysis, Spring 2015 (TA),

MATE-1105 Linear Algebra, Fall 2012, Spring 2013, Fall 2014 (TA),

MATE-3410 Differential Geometry Fall 2013 (TA),

MATE-1214 Integral Calculus Fall 2013 (TA),

ISIS-1206 Data Structures Spring 2012 (TA).

Skills

Programming: C++, Python, Julia, Rust, Matlab, LaTeX.
Languages: Fluent in English, Portuguese, and Spanish (native).

Last updated: November 3, 2026